

Danling Wei

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Summary

AI Engineer / Software Engineer with 2+ years of experience building production LLM applications, RAG systems, agentic workflows, and data pipelines for enterprise content and product platforms. Strong in Python, Agent Orchestration, retrieval, LLM evaluation, Elasticsearch, Redis, Docker, and backend API development. Columbia MS graduate with end-to-end ownership across architecture, implementation, quality evaluation, and production rollout.

Work Experience

PiSrc

New York, NY

Software Engineer

Mar 2024 – Present

AI Skills Orchestrator – Agentic Authoring & QA Agents

- Architected and shipped a tool-calling Skills Orchestrator for AEM/CMS authoring and QA; decomposed content requests into Planner, Executor, and QA agent flows with schema-bound skills, reducing repetitive authoring time by 60%+.
- Built validation and evaluation gates for page structure, metadata completeness, multilingual consistency, and output format correctness; logged skill inputs, outputs, latency, and failures to support traceability and quality analysis.
- Productionized CMS and LLM API integrations with rate limiting, retries, fallbacks, structured output validation, and human-review escalation to prevent malformed LLM output from reaching production content.

Agentic RAG Chatbot Optimization

- Upgraded a production RAG customer-support chatbot from general-purpose retrieval to an agentic retrieval workflow with intent detection, index routing, LLM-based query rewriting, hybrid retrieval, and reranking.
- Designed an intent-detection layer to classify user queries and route them to the most relevant index, replacing broad retrieval with index-specific search and higher-quality rewritten queries.
- Improved document ingestion and indexing quality with Docling-based PDF processing, table-aware chunk enrichment, and HTML H1/H2/H3 hierarchy detection to preserve document context and improve retrieval precision.
- Tuned retrieval relevance with offline evaluation on representative support queries; improved retrieval accuracy by ~40% and user satisfaction by ~60%.

Product Data Integration Pipeline – LLM-powered Translation & Processing

- Designed an LLM-assisted data pipeline ingesting 500K+ product records and 100K+ daily deltas from four heterogeneous APIs into Elasticsearch, standardizing multilingual product metadata for global search.
- Built Dagster-based batch and incremental workflows with idempotent processing, schema evolution, backfill and repair paths, validation checks, failure recovery, and observability.
- Redesigned Elasticsearch indexing strategy and data processing flow, improving query performance by 3x and reducing manual operations by ~40%.
- Automated multilingual product translation with an LLM-assisted pipeline, replacing manual translation workflows and contributing to ~\$500K estimated annual translation cost savings

Skills

Languages: Python, Java, TypeScript, JavaScript, Shell

AI / LLM Systems: RAG, Agentic Workflows, Tool Calling, LLM Evaluation, Prompt Engineering

Retrieval & Data: Elasticsearch, Weaviate, Redis, MySQL, Dagster, Hybrid Retrieval

Backend & Infrastructure: FastAPI, REST APIs, Docker, CI/CD, GitHub Actions, Azure

Quality & Observability: pytest, Unit Testing, Integration Testing, Retrieval Evaluation

Education

Columbia University

New York, United States

Master of Science, Electrical Engineering (Computer Science Track)

Sep 2022 – Dec 2023

Beijing University of Technology

Beijing, China

Bachelor of Science in Software Engineering

Sep 2018 – Jun 2022